VANISHING OF COHOMOLOGY ABOVE THE DIMENSION OF A QUASI-COMPACT QUASI-SEPARATED ALGEBRAIC SPACE, WITH SURJECTIVITY IN THE TOP DEGREE AND VANISHING WITH SUPPORT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Dimension is invariant under etale morphisms, and for a quasi-separated decent space it equals the Krull dimension. Properties of Spaces, label spaces-properties-lemma-dimension-decent-invariant-under-etale (invoked at proof line 1785) and Decent Spaces, label decent-spaces-lemma-dimension-decent-space (invoked at proof line 1789). They let the argument work with honest Krull dimension on the auxiliary schemes that appear, everything etale over X having dimension at most d.
- Etale cohomology of a quasi-coherent module on a scheme agrees with usual cohomology, also with support in a closed subscheme. Descent, label descent-proposition-same-cohomology-quasi-coherent (invoked at proof line 1795) and Etale Cohomology, label etale-cohomology-lemma-cohomology-with-support-quasi-coherent (invoked at proof line 1798). They identify the cohomology computed on the auxiliary schemes V_p with the usual sheaf cohomology, where the vanishing statement for schemes applies.
- Filtration of a quasi-compact quasi-separated algebraic space by open pieces with etale schemes over them. Decent Spaces, label decent-spaces-lemma-filter-quasi-compact-quasi-separated (invoked at proof line 1803). It provides the chain $\emptyset = U_{n+1} \subset U_n \subset \ldots \subset U_1 = X$ and the quasi-compact separated schemes $V_p \to U_p$ which are surjective etale and isomorphisms over $T_p = U_p \setminus U_{p+1}$.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Vanishing of cohomology above the cohomological dimension of a scheme. Cohomology, label cohomology-proposition-cohomological-dimension-spectral (invoked at proof lines 1818 and 1831). It gives the vanishing over the auxiliary schemes V_p, which is transported back along the filtration pieces.
- Localization of cohomology with support along an etale morphism that is an isomorphism over the support. Same chapter spaces-cohomology.tex, label lemma-etale-localization-sheaf-with-support (invoked at proof lines 1828 and 1839). It converts the cohomology with support $H^q_{T_p}(U_p, \mathcal{F})$ into cohomology with support on the scheme V_p, where vanishing is known.

Source reference. Original proposition 0A3G: sources/stacks-chow/tags.txt, source lines 12739–12739. Navigation pointer for \ref{cohomology-proposition-cohomological-dimension-spectral} used inside the counted span (line 1818); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A4J: sources/stacks-chow/tags.txt, source lines 12777–12777. Navigation pointer for \ref{decent-spaces-lemma-dimension-decent-space} used inside the counted span (line 1789); the argument that consumes it is retained in the span.

Source reference. Original lemma 07ST: sources/stacks-chow/tags.txt, source lines 9913–9913. Navigation pointer for \ref{decent-spaces-lemma-filter-quasi-compact-quasi-separated} used inside the counted span (line 1803); the argument that consumes it is retained in the span.

Source reference. Original proposition 03DW: sources/stacks-chow/tags.txt, source lines 4435–4435. Navigation pointer for \ref{descent-proposition-same-cohomology-quasi-coherent} used inside the counted span (line 1795); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A46: sources/stacks-chow/tags.txt, source lines 12764–12764. Navigation pointer for \ref{etale-cohomology-lemma-cohomology-with-support-quasi-coherent} used inside the counted span (line 1798); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A4P: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1674–1674. Navigation pointer for \ref{lemma-etale-localization-sheaf-with-support} used inside the counted span (line 1828); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A4H: sources/stacks-chow/tags.txt, source lines 12775–12775. Navigation pointer for \ref{spaces-properties-lemma-dimension-decent-invariant-under-etale} used inside the counted span (line 1785); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Assume $\dim(X) \leq d$ for some integer d. Let $\mathcal{F}$ be a quasi-coherent sheaf $\mathcal{F}$ on X.*

(1) *$H^q(X, \mathcal{F}) = 0$ for $q > d$,*
(2) *$H^d(X, \mathcal{F}) \to H^d(U, \mathcal{F})$ is surjective for any quasi-compact open $U \subset X$,*
(3) *$H^q_Z(X, \mathcal{F}) = 0$ for $q > d$ for any closed subspace $Z \subset X$ whose complement is quasi-compact.*

Proof. By Properties of Spaces, Lemma 0A4H every algebraic space Y étale over X has dimension $\leq d$. If Y is quasi-separated, the dimension of Y is equal to the Krull dimension of $|Y|$ by Decent Spaces, Lemma 0A4J. Also, if Y is a scheme,

then étale cohomology of $\mathcal{F}$ over Y, resp. étale cohomology of $\mathcal{F}$ with support in a closed subscheme, agrees with usual cohomology of $\mathcal{F}$, resp. usual cohomology with support in the closed subscheme. See Descent, Proposition 03DW and Étale Cohomology, Lemma 0A46. We will use these facts without further mention.

By Decent Spaces, Lemma 07ST there exist an integer n and open subspaces

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

with the following property: setting $T_p = U_p \setminus U_{p+1}$ (with reduced induced subspace structure) there exists a quasi-compact separated scheme V_p and a surjective étale morphism $f_p : V_p \to U_p$ such that $f_p^{-1}(T_p) \to T_p$ is an isomorphism.

As $U_n = V_n$ is a scheme, our initial remarks imply the cohomology of $\mathcal{F}$ over U_n vanishes in degrees $> d$ by Cohomology, Proposition 0A3G. Suppose we have shown, by induction, that $H^q(U_{p+1}, \mathcal{F}|_{U_{p+1}}) = 0$ for $q > d$. It suffices to show $H^q_{T_p}(U_p, \mathcal{F})$ for $q > d$ is zero in order to conclude the vanishing of cohomology of $\mathcal{F}$ over U_p in degrees $> d$. However, we have

$$H^q_{T_p}(U_p, \mathcal{F}) = H^q_{f_p^{-1}(T_p)}(V_p, \mathcal{F})$$

by Lemma 0A4P and as V_p is a scheme we obtain the desired vanishing from Cohomology, Proposition 0A3G. In this way we conclude that (1) is true.

To prove (2) let $U \subset X$ be a quasi-compact open subspace. Consider the open subspace $U' = U \cup U_n$. Let $Z = U' \setminus U$. Then $g : U_n \to U'$ is an étale morphism such that $g^{-1}(Z) \to Z$ is an isomorphism. Hence by Lemma 0A4P we have $H^q_Z(U', \mathcal{F}) = H^q_Z(U_n, \mathcal{F})$ which vanishes in degree $> d$ because U_n is a scheme and we can apply Cohomology, Proposition 0A3G. We conclude that $H^d(U', \mathcal{F}) \to H^d(U, \mathcal{F})$ is surjective. Assume, by induction, that we have reduced our problem to the case where U contains U_{p+1}. Then we set $U' = U \cup U_p$, set $Z = U' \setminus U$, and we argue using the morphism $f_p : V_p \to U'$ which is étale and has the property that $f_p^{-1}(Z) \to Z$ is an isomorphism. In other words, we again see that

$$H^q_Z(U', \mathcal{F}) = H^q_{f_p^{-1}(Z)}(V_p, \mathcal{F})$$

and we again see this vanishes in degrees $> d$. We conclude that $H^d(U', \mathcal{F}) \to H^d(U, \mathcal{F})$ is surjective. Eventually we reach the stage where $U_1 = X \subset U$ which finishes the proof.

A formal argument shows that (2) implies (3). □